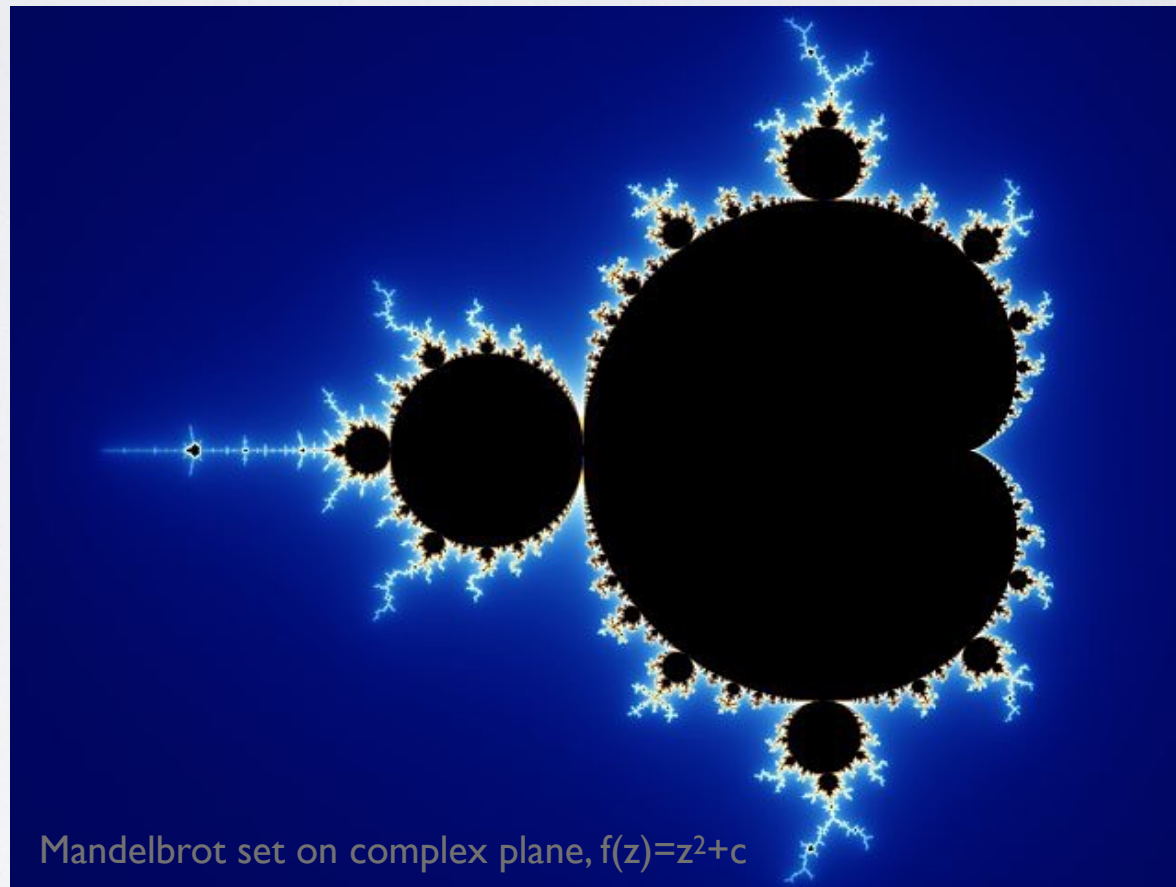
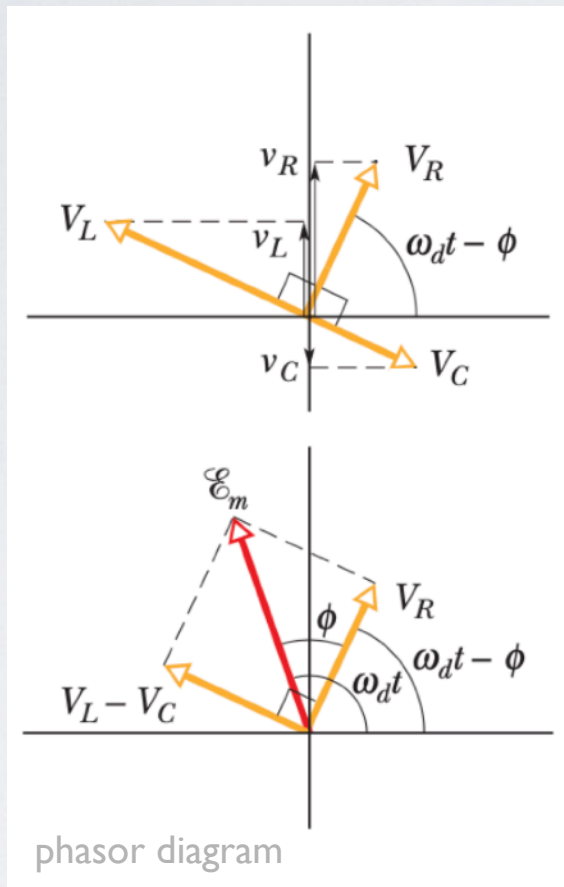


Week 2 - #2

Complex Numbers (II)



Today: Ch 1-2

Next Class: Ch 2

Ji-hoon Kim (Seoul National University)

Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #2

— [open book and open note, “and” cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Define “complex plane” from your textbook; then, Boas, Chapter 2, Section 5, Problem 59
2. Briefly explain the so-called Mandelbrot set demonstrated in the following movie — in particular, how this one of the most famous images in the fractal/chaos theory is plotted on a “complex plane”.



youtube.com/watch?v=G_GBwuYuOOs,
Mandelbrot set on complex plane

Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 2, Boas, 3rd ed.

The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

- Section 04: Problems 4, 13
- Section 05: Problems 6, 7, 27, 28, 47, 50, 57
- Section 06: Problems 11, 14
- Section 07: Problems 12, 13, 17
- Section 08: Problems 3
- Section 09: Problems 9, 24, 25, 27, 28, 37
- Section 10: Problems 24, 32, 33
- Section 11: Problems 18
- Section 12: Problems 1, 10, 14, 15, 27, 38
- Section 14: Problems 2, 7, 9, 24, 25
- Section 15: Problems 6, 7, 18
- Section 16: Problems 1, 2, 5, 8, 11
- Section 17: Problems 19, 21, 25, 30

48 problems / 337 total

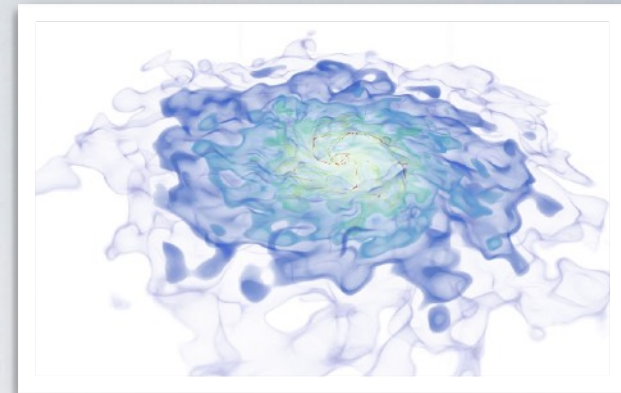
Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 3, Boas, 3rd ed.

The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

- Section 02: Problems 8, 13, 14, 17, 18
- Section 03: Problems 2, 6, 9, 10, 13, 15
- Section 04: Problems 5, 6, 9, 21, 23
- Section 05: Problems 17, 21, 32, 35, 37, 38
- Section 06: Problems 6, 7, 16, 21, 29, 30
- Section 07: Problems 12, 27, 31, 34, 35
- Section 08: Problems 2, 10, 15, 16, 17, 24
- Section 09: Problems 3, 5, 10, 17, 19(c), 24, 25(a)(b)
- Section 10: Problems 4(c), 5(a), 7
- Section 11: Problems 9, 10, 19, 21, 31, 33, 42, 44, 50, 57, 60, 61
- Section 12: Problems 9 (for Problems 4, 7), 16, 21
- Section 13: Problems 1, 4, 7
- Section 14: Problems 13

Chapter 10, Sections 1-4

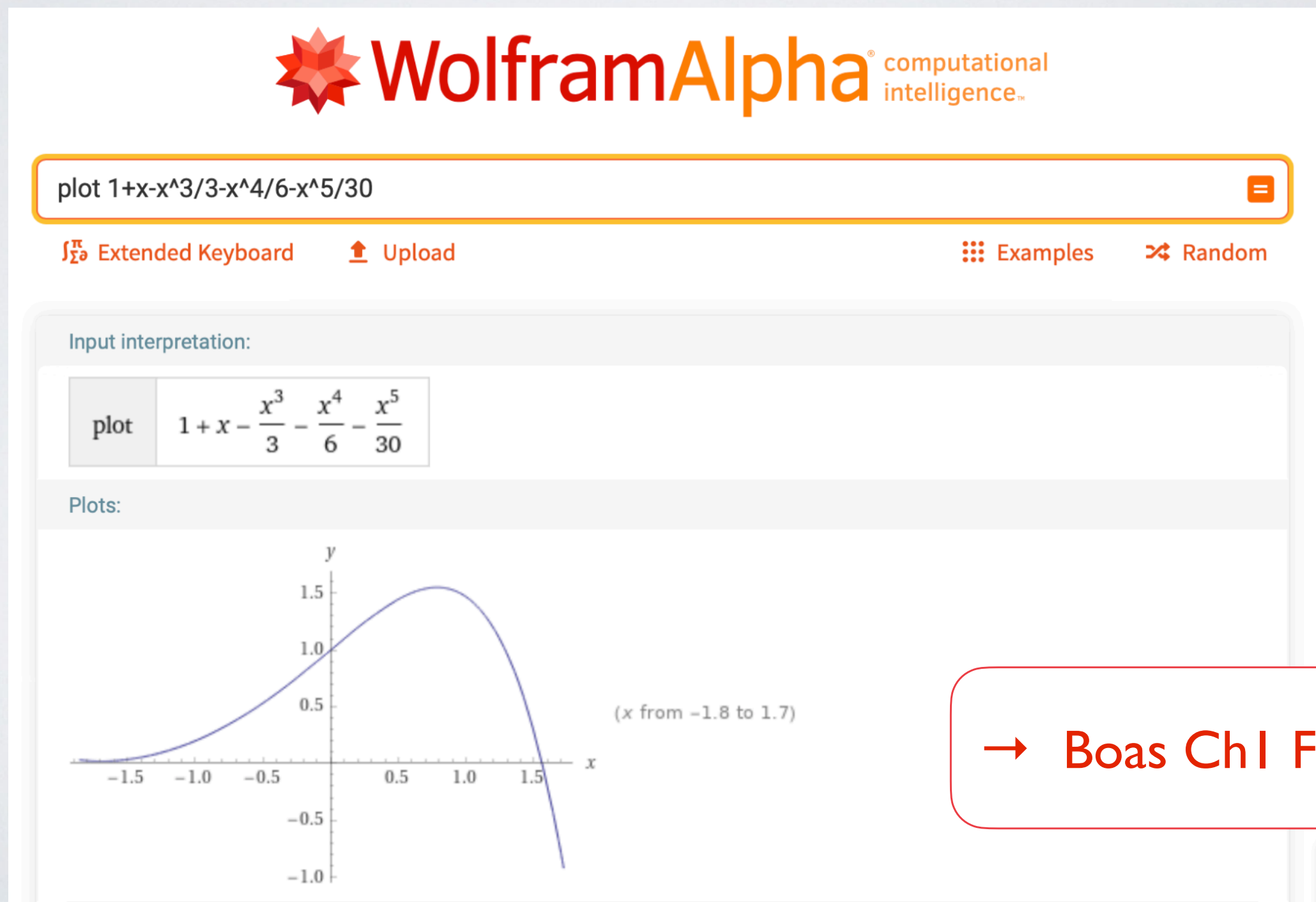
- Section 02: Problems 6
- Section 03: Problems 1, 2, 8
- Section 04: Problems 2, 6, 8



Taylor Series Using A Computer

Numerical Problems

- Tools you may want to try for your numerical problems:
Matlab, Mathematica, Mathcad, **Wolfram Alpha**, Python, etc.



Taylor Series Using A Computer



taylor expand $e^x \cos x$



Extended Keyboard

Upload

Examples

Random

Input interpretation:

series

$e^x \cos(x)$

Series expansion at $x = 0$:

[Fewer terms](#)

[More terms](#)

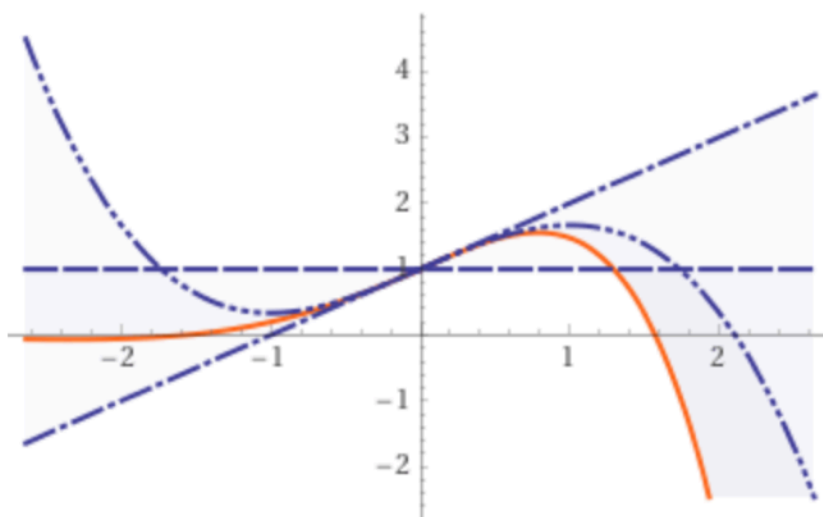
$$1 + x - \frac{x^3}{3} - \frac{x^4}{6} - \frac{x^5}{30} + \frac{x^7}{630} + \frac{x^8}{2520} + \frac{x^9}{22680} - \frac{x^{11}}{1247400} + O(x^{12})$$

(Taylor series)

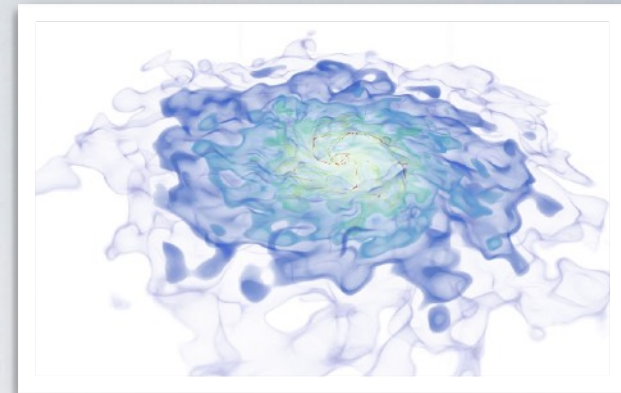
(converges everywhere)

Approximations about $x = 0$ up to order 3:

[...](#)



(order n approximation shown with n dots)



Pitfalls of Blind Computation

Pitfalls of Blind Computation

- Inaccurate estimation on Google:

The screenshot shows a Google search for the expression $\ln(\sqrt{(1+0.0015)/(1-0.0015)}) - \tan(0.0015)$. The search bar contains the expression, and the results show "About 602 results (0.47 seconds)". Below the results, a calculator interface displays the same expression as $\ln(\sqrt{(1 + 0.0015) / (1 - 0.0015)}) - \tan(0.0015 \text{ radians}) =$ followed by the result $\underline{6.0606901e-16}$. The calculator interface includes buttons for Rad, Deg, x!, (,), %, AC, Inv, sin, ln, 7, 8, 9, ÷, π, cos, log, 4, 5, 6, ×, e, tan, √, 1, 2, 3, −, Ans, EXP, x^y, 0, ., =, and +.

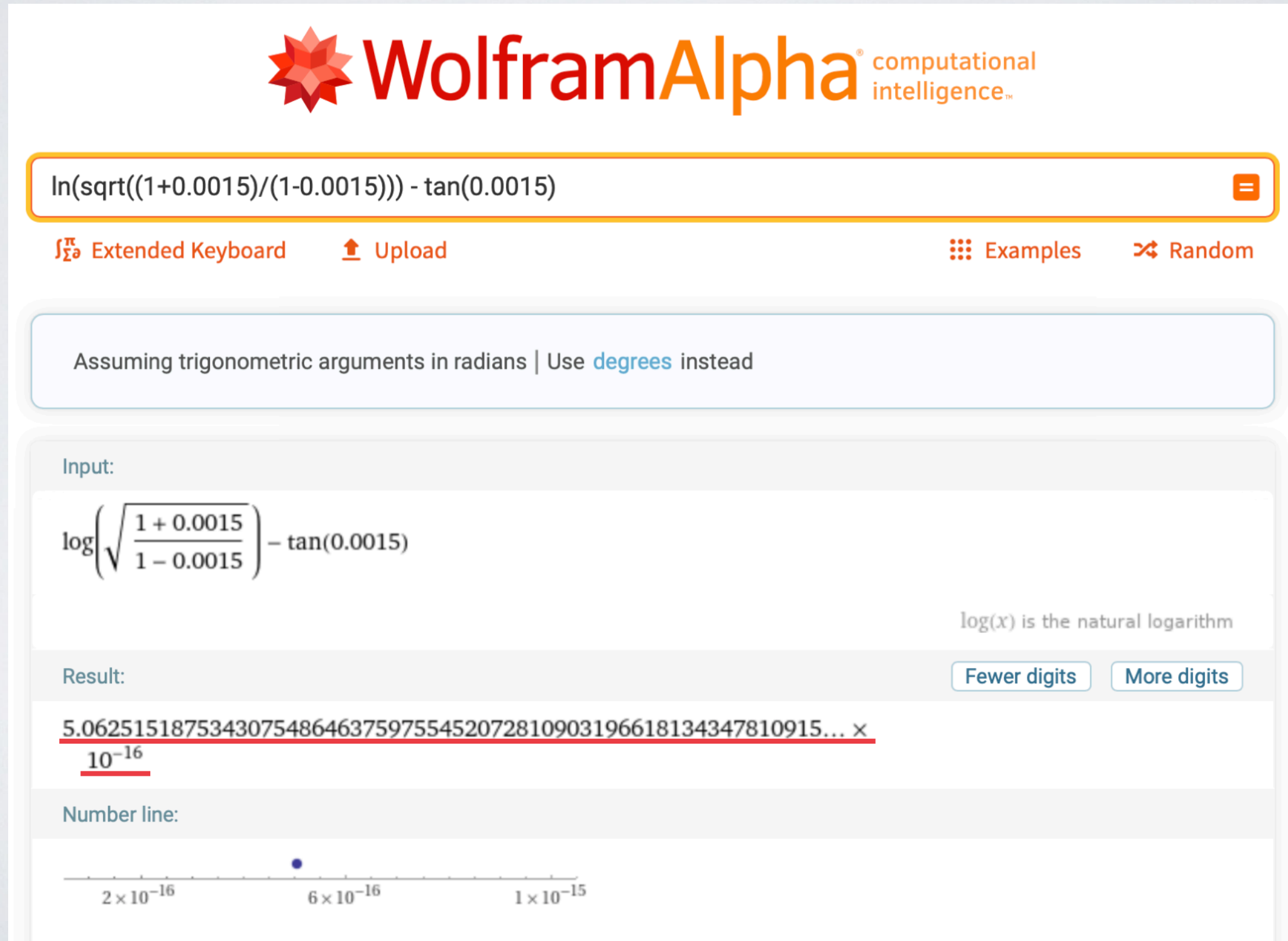
Pitfalls of Blind Computation

- Inaccurate estimation on Python:

```
bash-3.2$ python
Python 2.7.9 (default, Sep  9 2015, 22:07:15)
[GCC 4.2.1 Compatible Apple LLVM 6.1.0 (clang-602.0.53)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import math
>>> math.log(math.sqrt((1+0.0015)/(1-0.0015))) - math.tan(0.0015)
6.06069014419397e-16
```

Pitfalls of Blind Computation

- Accurate estimation on WolframAlpha:



The screenshot shows the WolframAlpha interface. At the top is the logo with a red star and the text "WolframAlpha computational intelligence". Below the logo is a search bar containing the expression $\ln(\sqrt{(1+0.0015)/(1-0.0015)}) - \tan(0.0015)$. To the right of the search bar are links for "Extended Keyboard", "Upload", "Examples", and "Random". Below the search bar is a light blue box stating "Assuming trigonometric arguments in radians | Use [degrees](#) instead". The main content area shows the input expression in a larger font: $\log\left(\sqrt{\frac{1+0.0015}{1-0.0015}}\right) - \tan(0.0015)$. To the right of this is a note: "log(x) is the natural logarithm". Below the input is the result: $5.0625151875343075486463759755452072810903196618134347810915... \times 10^{-16}$. The result is followed by two buttons: "Fewer digits" and "More digits". At the bottom is a "Number line" showing a scale from 2×10^{-16} to 1×10^{-15} , with a blue dot indicating the position of the result.

WolframAlpha[®] computational intelligence™

$\ln(\sqrt{(1+0.0015)/(1-0.0015)}) - \tan(0.0015)$

[Extended Keyboard](#) [Upload](#) [Examples](#) [Random](#)

Assuming trigonometric arguments in radians | Use [degrees](#) instead

Input:

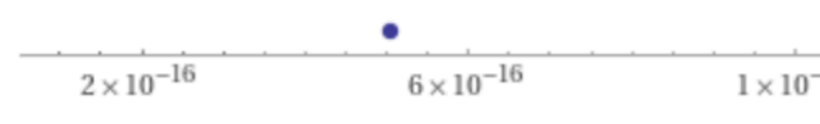
$$\log\left(\sqrt{\frac{1+0.0015}{1-0.0015}}\right) - \tan(0.0015)$$

log(x) is the natural logarithm

Result: [Fewer digits](#) [More digits](#)

$5.0625151875343075486463759755452072810903196618134347810915... \times 10^{-16}$

Number line:



The number line shows a scale from 2×10^{-16} to 1×10^{-15} . A blue dot is placed on the line, indicating the position of the result.

Pitfalls of Blind Computation

- Accurate estimation on Full Precision Calculator:

Full Precision Calculator

This calculator calculates answers to full* accuracy.
If you want more functions (but not full precision) try the [Scientific Calculator](#)

$\ln(\sqrt{(1+0.0015)/(1-0.0015)}) - \tan(0.0015 \cdot 180/\pi)$

**0.0000000000000000506251518753430754864637597554
5207281090319661813434781091560762983344871629
59393151147072175618546084275046494927908515814
2080150864641203183784359975199564279750604123
6612968728336675572119291634**

7	8	9	×	÷
4	5	6	+	-
1	2	3	(	)
0	.	^	←	C

deg

! e pi

History:

$\ln(\sqrt{(1+0.0015)/(1-0.0015)}) - \tan(0.0015 \cdot 180/\pi) =$
0.000000000000000050625151875343075486463759755452072810903
1966181343478109156076298334487162959393151147072175618546
0842750464949279085158142080150864641203183784359975199564
2797506041236612968728336675572119291634
 $\ln(\sqrt{(1+0.0015)/(1-0.0015)}) =$
0.001500001125001518752440852485777952715102659838705448085
5820599653054716307555148275394381298635589258005404247102

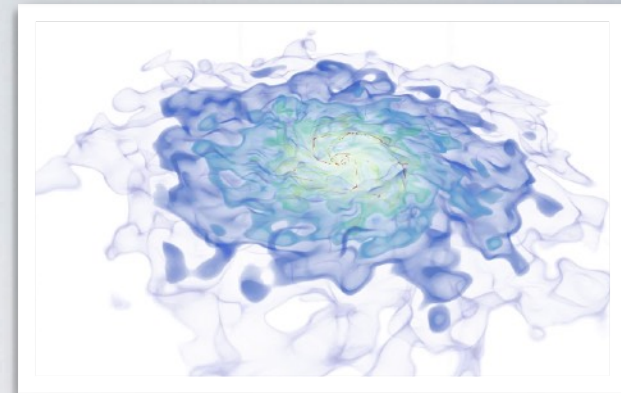
Decimals: **200**

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Pitfalls of Blind Computation

- Inaccurate estimation on Python:

```
bash-3.2$ python
Python 2.7.9 (default, Sep  9 2015, 22:07:15)
[GCC 4.2.1 Compatible Apple LLVM 6.1.0 (clang-602.0.53)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import math
>>> math.log(math.sqrt((1+0.0015)/(1-0.0015))) - math.tan(0.0015)
6.06069014419397e-16
>>> math.log(math.sqrt((1+0.0015)/(1-0.0015)))
0.0015000011250016186
>>> math.tan(0.0015)
0.0015000011250010125
>>>
>>>
```



Pitfalls of Blind Computation (Another Example: Problem 15.2)

Pitfalls of Blind Computation

- Inaccurate estimation on Google:

The screenshot shows a Google search for the expression $1/\sqrt{1+0.012^4}-\cos(0.012^2)$. The search bar contains the expression, and the results show "About 226,000 results (0.55 seconds)". Below the results, a calculator interface is displayed. The calculator shows the expression $(1 / \sqrt{1 + (0.012^4)}) - \cos(0.012^2 \text{ radians}) =$ and the result 0 . The calculator interface includes buttons for Rad, Deg, x!, (,), %, AC, Inv, sin, ln, 7, 8, 9, ÷, π, cos, log, 4, 5, 6, ×, e, tan, √, 1, 2, 3, −, Ans, EXP, x^y, 0, ., =, and +.

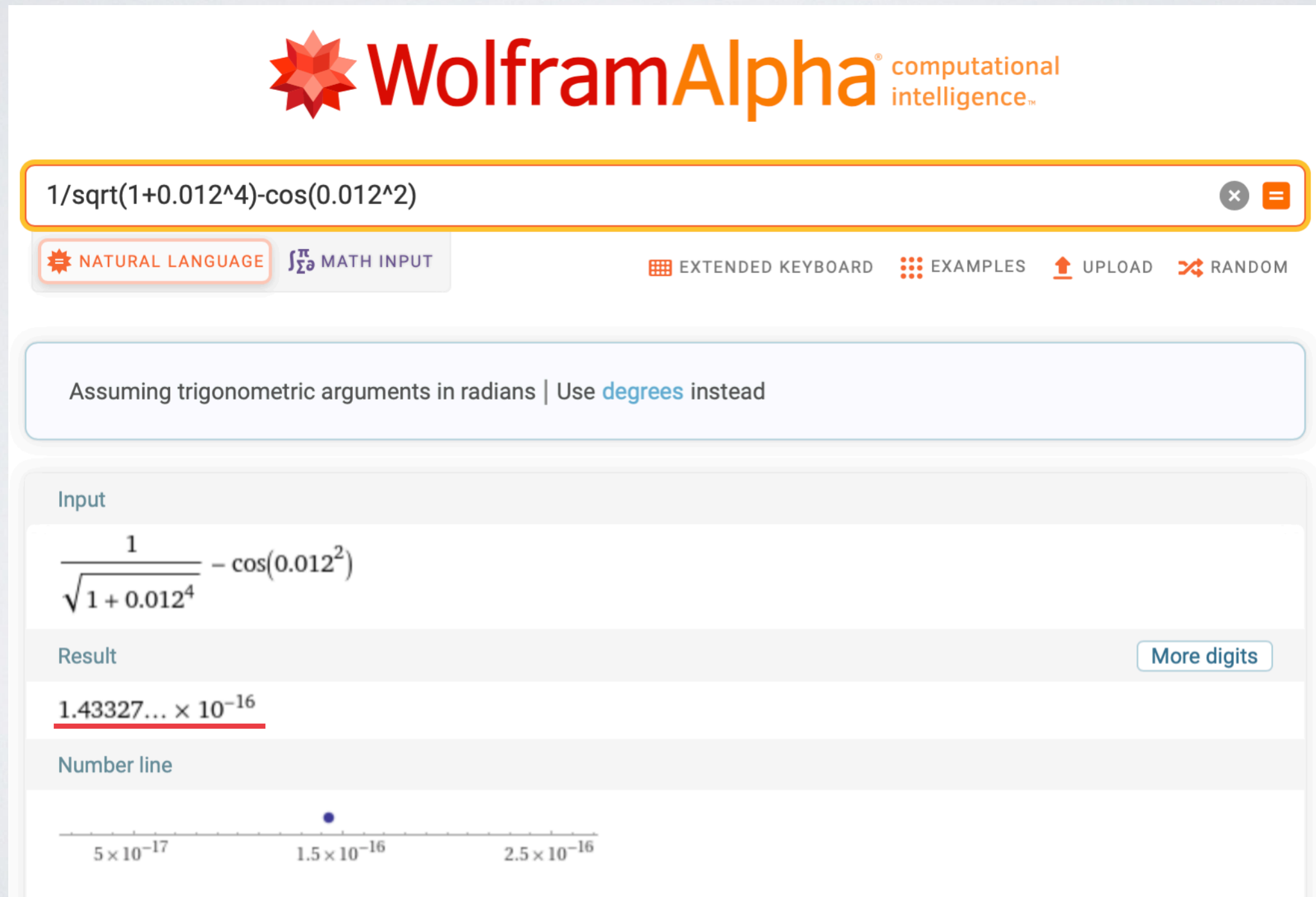
Pitfalls of Blind Computation

- Inaccurate estimation on Python:

```
bash-3.2$ python
Python 2.7.9 (default, Sep  9 2015, 22:07:15)
[GCC 4.2.1 Compatible Apple LLVM 6.1.0 (clang-602.0.53)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import math
>>> 1/math.sqrt(1+0.012**4) - math.cos(0.012**2)
1.1102230246251565e-16
>>>
```

Pitfalls of Blind Computation

- Accurate estimation on WolframAlpha:



The screenshot shows the WolframAlpha interface. At the top is the logo with the text "WolframAlpha computational intelligence". Below it is a search bar containing the expression $1/\sqrt{1+0.012^4}-\cos(0.012^2)$. Under the search bar are two tabs: "NATURAL LANGUAGE" and "MATH INPUT", with "MATH INPUT" being selected. To the right of the tabs are links for "EXTENDED KEYBOARD", "EXAMPLES", "UPLOAD", and "RANDOM". Below the search bar is a message: "Assuming trigonometric arguments in radians | Use [degrees](#) instead". The main content area is divided into sections: "Input" showing the expression in a large font, "Result" showing the value $1.43327... \times 10^{-16}$ with a "More digits" button, and "Number line" showing a horizontal axis with a point at 1.5×10^{-16} .

WolframAlpha[®] computational intelligence™

$1/\sqrt{1+0.012^4}-\cos(0.012^2)$

NATURAL LANGUAGE MATH INPUT

EXTENDED KEYBOARD EXAMPLES UPLOAD RANDOM

Assuming trigonometric arguments in radians | Use [degrees](#) instead

Input

$$\frac{1}{\sqrt{1+0.012^4}} - \cos(0.012^2)$$

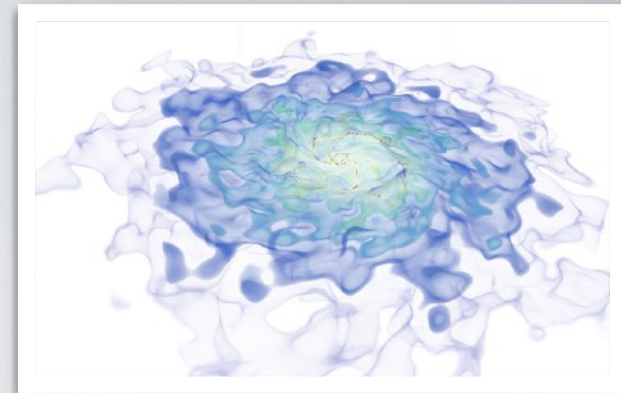
Result [More digits](#)

$1.43327... \times 10^{-16}$

Number line



5 × 10⁻¹⁷ 1.5 × 10⁻¹⁶ 2.5 × 10⁻¹⁶



Roots of Complex Numbers

Roots of Complex Numbers



$(-8i)^{(1/6)}$



NATURAL LANGUAGE

MATH INPUT

EXTENDED KEYBOARD

EXAMPLES

UPLOAD

RANDOM

Assuming i is the imaginary unit | Use i as a [variable](#) instead

Input

$\sqrt[6]{-8i}$

i is the imaginary unit

Exact result

$-(-1)^{11/12} \sqrt{2}$

Decimal approximation

Enlarge | Data | Customize | Plain Text

1.36602540378443864676372317075293618347140262690519031402790348...

-

0.366025403784438646763723170752936183471402626905190314027903489...

i

Polar coordinates

Radians ▾

Approximate forms

$r = \sqrt{2}$ (radius), $\theta = -\frac{\pi}{12}$ (angle)

Roots of Complex Numbers



$(-8i)^{(1/6)}$

NATURAL LANGUAGE

MATH INPUT

EXTENDED KEYBOARD

EXAMPLES

UPLOAD

RANDOM

All 6th roots of $-8i$

Fewer roots

More digits

Polar form

$$\sqrt{2} e^{-i\pi/12} \approx 1.36603 - 0.36603 i$$

$$\sqrt{2} e^{i\pi/4} = 1 + i \text{ (principal root)}$$

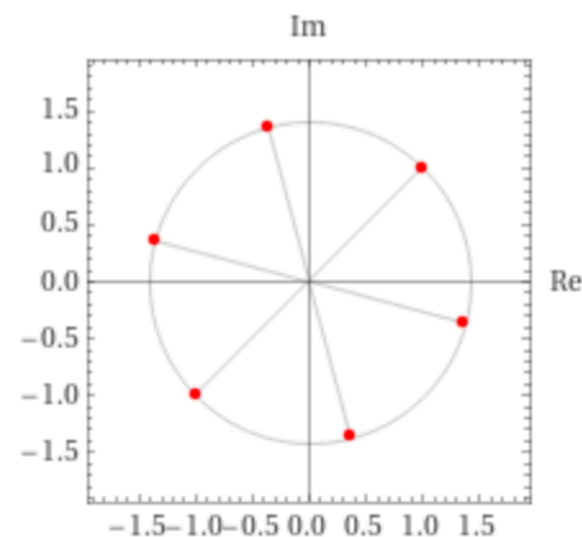
$$\sqrt{2} e^{7i\pi/12} \approx -0.3660 + 1.3660 i$$

$$\sqrt{2} e^{11i\pi/12} \approx -1.3660 + 0.3660 i$$

$$\sqrt{2} e^{-3i\pi/4} = -1 - i$$

$$\sqrt{2} e^{-5i\pi/12} \approx 0.3660 - 1.3660 i$$

Plot of all roots in the complex plane



→ Boas Ch2 Fig 10.4